

Keshara Jayasinghe | Software Engineer

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SUMMARY

Software engineer with 5+ years of experience building web, mobile, and embedded applications across the full stack. Front-end work centres on Angular and Flutter; back-end on Node.js and NestJS, with Firebase and Google Cloud Platform for infrastructure. Recent work includes an automated RAG pipeline over internal PostgreSQL data and a Python/Power BI reporting project. Comfortable owning features end to end, from Figma designs through to CI/CD deployment.

SKILLS

Languages: JavaScript, TypeScript, Python, Dart, C++, Go, Embedded C

Front end: Angular, React, Next.js, RxJS, Flutter, Ionic, HTML5, CSS3, SCSS

Back end: Node.js, NestJS

Cloud and databases: Google Cloud Platform (GCP), AWS, Firebase, Cloud SQL (PostgreSQL), SQL, NoSQL

DevOps and tools: Git, GitHub Actions, Google Cloud Build, Docker, Jira, Notion

Testing: Jasmine, Jest, Pytest

Design: Figma, Adobe XD, Sketch

AI tooling: Claude Code, Google Antigravity, Google AI Studio

Other: Power BI, Microsoft Office 365

EXPERIENCE

Freelance Software Developer | Australia | May 2024 – Present

- Architected a cross-platform fleet and property management suite using Flutter and Angular, supporting asset tracking for a growing user base.
- Built a role-based access control (RBAC) system with Firebase Authentication to secure data across mobile and web interfaces.
- Developed a full-stack booking management system with Next.js and NestJS; its automated admin portal cut reservation workflow effort by 90%.
- Designed a relational schema in Cloud SQL (PostgreSQL) handling booking states, room availability logic, and transaction integrity.
- Built an automated RAG pipeline with n8n and Python so LLMs could query internal PostgreSQL databases; added vector embeddings and semantic search to reduce hallucination rates in customer-facing workflows.
- Delivered a hospitality platform front end in Qwik with near-instant load times and 100/100 Google Lighthouse SEO scores.

Software Engineer, YellowZap | Dubai, UAE | September 2021 – February 2024

- Developed responsive e-commerce web applications in Angular; used server-side rendering with Angular Universal to cut load times by 30% and improved SEO and Lighthouse scores.
- Built reusable Angular UI components to standardise development and speed up delivery, working from Figma designs in HTML5 and SCSS.
- Integrated APIs, payment gateways, OTP authentication, Google Maps, Google Analytics, Tag Manager, and Microsoft Clarity in collaboration with back-end developers and designers.
- Wrote geo-location algorithms for regional service allocation and Node.js cloud functions for email sending, image uploads, and automatic payment capture.
- Set up CI/CD pipelines with GitHub Actions, Google Cloud Build, and Docker, reducing deployment times by over 95%.
- Maintained code quality with front-end and back-end unit tests in Jasmine and Jest.

Software Engineer, Orel IT | Colombo, Sri Lanka | *September 2020 – September 2021*

- Built responsive online store applications and admin panels in Angular with product search, sorting, discounts, promotions, and delivery tracking.
- Created Flutter apps for warranty activation, claims, invoicing, and warehouse stock counting, cutting processing time by 80% (5 days to 1); features included multi-user roles, OTP authentication, QR/barcode scanning, and geo-location.
- Added localisation and integrated Firebase, AWS, and geo-location services for real-time data.
- Published apps to TestFlight, the App Store, and Google Play.

Freelance Software Developer | Sri Lanka | *November 2019 – September 2020*

- Delivered web and mobile applications with Angular, Flutter, and Ionic, including e-commerce platforms, admin panels, and monitoring tools for embedded systems.
- Designed responsive UI/UX in Figma and Adobe XD; integrated AWS and Firebase (Firestore and Realtime Database) with embedded systems for real-time monitoring.

Embedded Systems Engineer, Nikko Innovations | Sri Lanka | *September 2018 – October 2019*

- Developed embedded software and hardware for healthcare intercom systems, autonomous surveillance robots (Raspberry Pi, ROS, Python), and real-time image processing systems (OpenCV, Python).
- Ran simulations for mobile robotic platforms and sensors; integrated AWS with IoT devices for Alexa voice control.

Electronics Engineer Intern, Orel Corporation | Colombo, Sri Lanka | *February 2017 – February 2018*

- Optimised IoT device software for power saving, fail-safe operation, and network security.
- Developed RF mesh network simulation software in C#; integrated AWS for Alexa voice control.

PROJECTS

FedLibrary Dashboard Solution | Federation University, Australia

MTech (Software Engineering) Group Project | *February 2025 – November 2025*

- Built an automated Python pipeline pulling resource usage data from the eReserve+ API into MS SQL Server alongside student performance data, giving academic and library staff real-time insights
- Created Power BI dashboards to visualise the combined datasets, automating report generation and cutting manual spreadsheet work
- Technologies: Python, Power BI, MS SQL Server, REST APIs.

EDUCATION

Master of Technology (Software Engineering)

Federation University, Australia | 2025 | GPA: 6.68/7.0

Bachelor of Engineering (Hons) – Mechatronics Engineering

University of Wolverhampton, UK | 2019

CERTIFICATIONS

- Cloud Architecture with Google Cloud Specialization
- Architecting with Google Compute Engine Specialization
- Introduction to Artificial Intelligence